CSSEMA 384 – Data Mining with Programming
Rose-Hulman Institute of Technology

Worksheet 01: Exploring Data

Name (Print):_____ Section:_____

Topic: Exploring Data

Work with a neighbor. Circle or select answers as directed. Fill only the short blanks; no coding is required.

1. **Explore Gapminder.** Open `https://www.gapminder.org/tools/` or follow the class demonstration. Use the income–life expectancy bubble chart, with population as size and world region as color. Pause at one year: _____.

   Circle one answer per line:

   (a) One bubble represents a: country / person / course.
   (b) Income per person is shown by: horizontal position / color / size.
   (c) Life expectancy is shown by: vertical position / color / size.
   (d) Population is shown by: color / size / vertical position.

2. **Behind the chart.** Imagine one row per country at the selected year. Circle *all* columns needed to reproduce the view and identify its time context:

   country name income per person life expectancy population

   world region year student ID favorite movie

   If we store every year, one row should represent a: country–year / person.

3. **Ask questions from data.** For each question, circle *Chart* if the displayed country data can answer it, or *More data* if additional evidence is needed.

   (a) Which countries have higher life expectancy in this year? Chart / More data
   (b) How did a country's income change across years? Chart / More data
   (c) Did higher income *cause* longer lives? Chart / More data

   Circle one question above that you would explore first; compare with a neighbor.

4. **Trace a source.** Find documentation for one indicator through the chart or `https://www.gapminder.org/data/`.

   Indicator: _____ Source organization: _____

   Circle the stronger source record:

   (a) "I found it on the internet."
   (b) Indicator name, definition/unit, source organization, and documentation link.

   Circle one item to check before comparing countries: units / missing values / both.

5. **Student records and data types.** A table has one row per student, with the features below. Circle one answer per line:

   A row represents a: student / feature. A column represents a: student / feature.

   Circle one type for each feature. Class year means first-year, sophomore, junior, or senior.

Feature	**Circle one type**
Student ID	Unordered category / Ordered category / Quantity
Major	Unordered category / Ordered category / Quantity
Class year	Unordered category / Ordered category / Quantity
Height (cm)	Unordered category / Ordered category / Quantity
Courses this term	Unordered category / Ordered category / Quantity

   Circle one: Number of courses is a *count / measurement*; height is a *count / measurement*.

6. **Meaning versus storage.** Circle one answer for each.

   (a) Averaging student IDs is meaningful: Yes / No.

   (b) Why? IDs measure an amount / IDs are labels.

   (c) Class year coded 1, 2, 3, 4 is automatically quantitative: Yes / No.

   (d) A meaningful summary of major is: mean major code / count in each major.

7. **The data-mining cycle.** Circle the sequence used in the legacy cycle:

   (a) Gather → clean → explore → model → interpret.

   (b) Model → interpret → gather → clean → explore.

   Looking for patterns in Gapminder is mainly: gathering / exploring / modeling.

8. **Reflection.** Circle one topic you found interesting and underline one to revisit:

   reading charts records/features data types numeric labels data sources

   Circle one helpful next step: another example / class discussion / more practice.